C. Balance

time limit per test: 3 seconds
memory limit per test: 128 megabytes

Nick likes strings very much, he likes to rotate them, sort them, rearrange characters within a string... Once he wrote a random string of characters a, b, c on a piece of paper and began to perform the following operations:

- to take two adjacent characters and replace the second character with the first one,
- to take two adjacent characters and replace the first character with the second one

To understand these actions better, let's take a look at a string «abc». All of the following strings can be obtained by performing one of the described operations on «abc»: «bbc», «abb», «acc». Let's denote the frequency of a character for each of the characters a, b and c as the number of occurrences of this character in the string. For example, for string «abc»: $|a| = 1$, $|b| = 1$, $|c| = 1$, and for string «bbc»: $|a| = 0$, $|b| = 2$, $|c| = 1$.

While performing the described operations, Nick sometimes got balanced strings. Let's say that a string is balanced, if the frequencies of each character differ by at most 1. That is $-1 \le |a| - |b| \le 1$, $-1 \le |a| - |c| \le 1$ и $-1 \le |b| - |c| \le 1$.

Would you help Nick find the number of different balanced strings that can be obtained by performing the operations described above, perhaps multiple times, on the given string s. This number should be calculated modulo 51123987.

Input

The first line contains integer n $(1 \le n \le 150)$ — the length of the given string s. Next line contains the given string s. The initial string can be balanced as well, in this case it should be counted too. The given string s consists only of characters a, b and c.

Output

Output the only number — the number of different balanced strings that can be obtained by performing the described operations, perhaps multiple times, on the given string s, modulo 51123987.

Examples

input Copy

```
4
abca
```

output Copy

```
7
```

input Copy

```
4
abbc
```

output Copy

```
3
```

input Copy

```
2
ab
```

output Copy

```
1
```

Note

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 17

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370518689	Apr/10/2026 12:13	Accepted

→ Problem tags

dp *2500

No tag edit access

In the first sample it is possible to get 51 different strings through the described operations, but only 7 of them are balanced: «abca», «bbca», «bcca», «bcaa», «abcc», «abbc», «aabc». In the second sample: «abbc», «aabc», «abcc». In the third sample there is only one balanced string — «ab» itself.

→ **Contest materials**

- Announcement
- Tutorial

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